MUTATION AND REPAIR MECHANISM

MUTATIONS:

- The genetic macromolecule DNA is highly stable with regard to its base composition and sequence. However, DNA is not totally exempt from gradual change.
- Mutation refers to **a change in the DNA structure of a gene**. The substances (chemicals) which can induce mutations are collectively known as mutagens.
- The changes that occur in DNA on mutation are reflected in replication, transcription and translation.

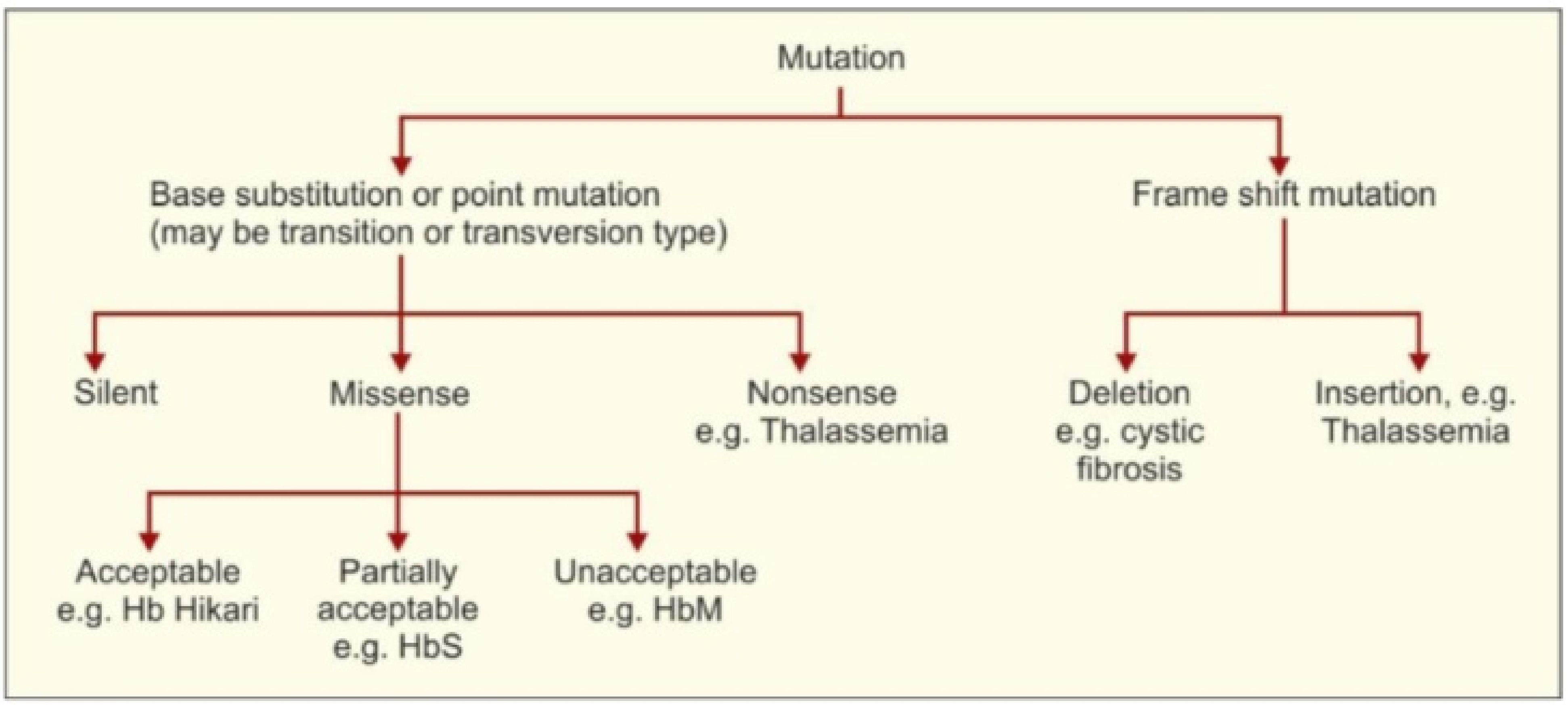

Figure 21.3: Types of mutation

TYPES OF MUTATIONS:

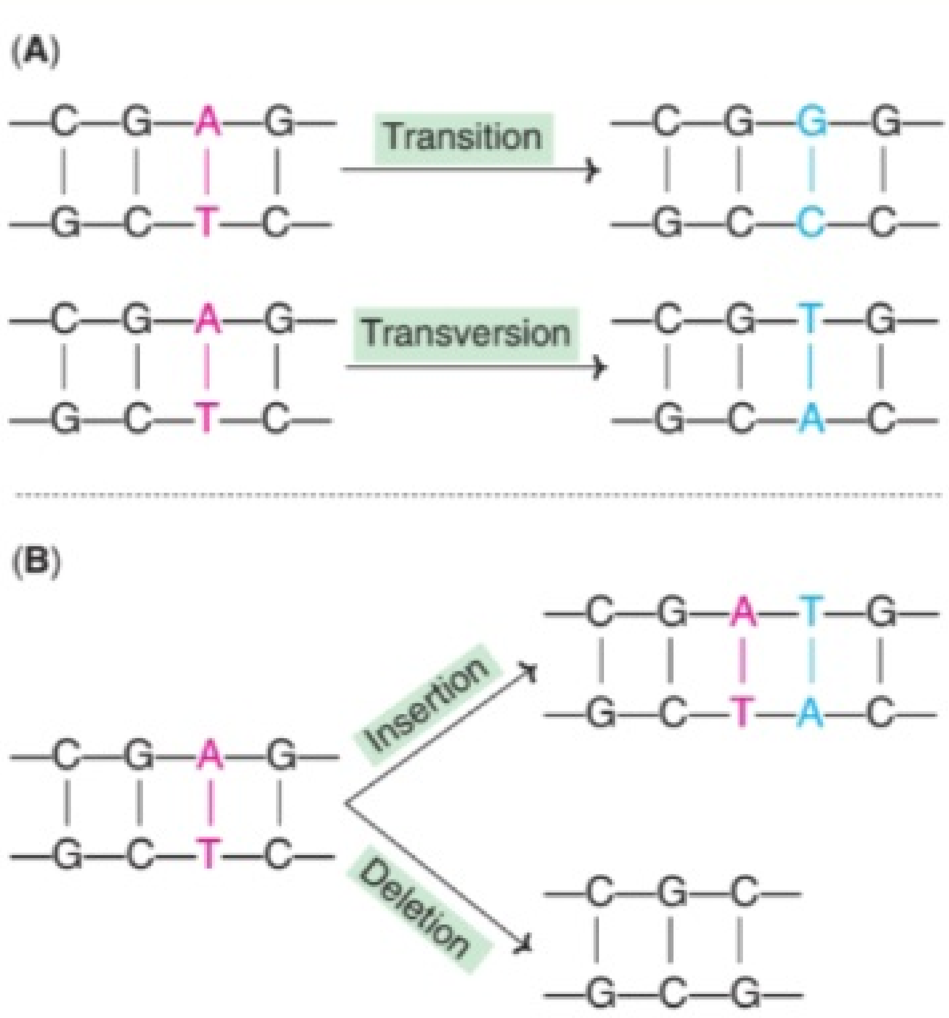

***Fig. 24.14 :** An illustration of mutations (**A**)-Point mutations; (**B**)-Frameshift mutations.*

Mutations are mainly of two major types:

1) Point mutations
2) Frame-shift mutations (Fig.24.14).

1) POINT MUTATIONS:

The replacement of one base pair by another, results in point mutation. They are of two sub-types.

(a) **Transitions:** In this case, a purine (or a pyrimidine) is replaced by another.

(b) **Transversions:** These are characterized by replacement of a purine by a pyrimidine or vice versa.

2) FRAMESHIFT MUTATIONS:

These occur when one or more base pairs are inserted in or deleted from the DNA, respectively, causing **insertion or deletion mutations**.

CONSEQUENCES OF POINT MUTATIONS:

The change in a single base sequence in point mutation may cause one of the following (Fig.24.15).

1) SILENT MUTATION:

The codon (of mRNA) containing the changed base may code for the same amino acid. For instance, UC**A** codes for serine and change in the third base (UC**U**) still codes for serine. This is due to degeneracy of the genetic code. Therefore, there are **no detectable effects** in silent mutation.

2) MISSENSE MUTATION:

In this case, the changed base may code for a different amino acid. For example, **U**CA codes for serine while **A**CA codes for threonine. The mistaken (or missense) amino acid may **be acceptable, partially acceptable or unacceptable** with regard to the function of protein molecule. **Sickle-cell anemia** is a classical example of missense mutation.

3) NONSENSE MUTATION:

Sometimes, the codon with the altered base may become a **termination** (or nonsense) **codon**. For instance, change in the second base of serine codon (UCA) may result in UAA. The altered codon acts as a stop signal and causes termination of protein synthesis, at that point.

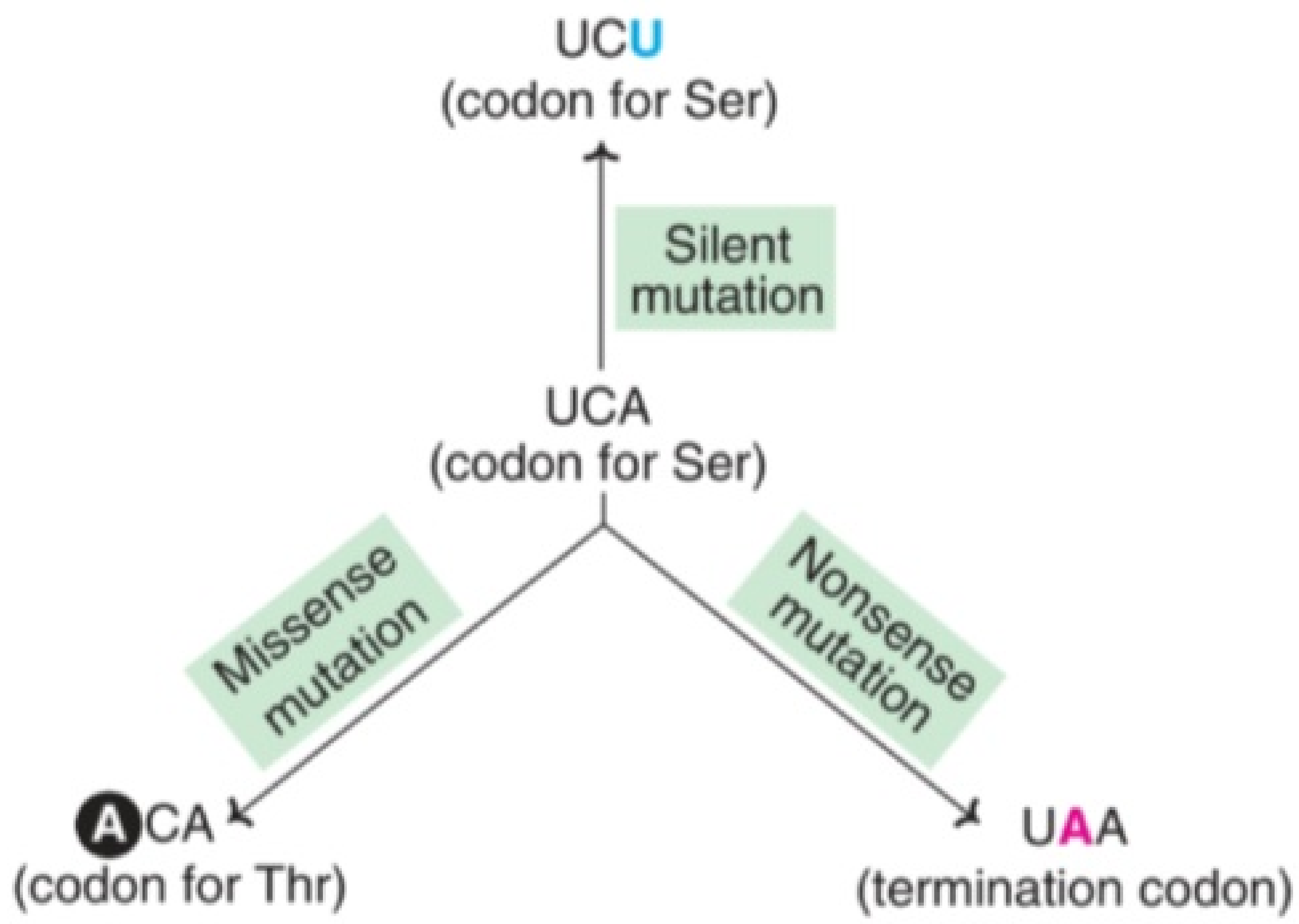

***Fig. 24.15 :** An illustration of point mutations (represented by a codon of mRNA).*

CONSEQUENCES OF FRAMESHIFT MUTATIONS:

The insertion or deletion of a base in a gene results in an **altered reading frame of the mRNA** (hence the name frameshift). The machinery of mRNA (containing codons) does not recognize that a base was missing or a new base was added. Since there are no punctuations in the reading of codons, translation continues. The result is that the protein synthesized will have several altered amino acids and/or prematurely terminated protein.

MUTATIONS AND CANCER:

Mutations are permanent alterations in DNA structure, which have been implicated in the etiopathogenesis of cancer.

REPAIR MECHANISMS:

As already stated, damage to DNA caused by replication errors or mutations may have serious consequences. The cell possesses an inbuilt system to repair the damaged DNA. This may be achieved by four distinct mechanisms (Table 24.2).

1) Base excision-repair
2) Nucleotide excision-repair
3) Mismatch repair
4) Double-strand break repair.

TABLE 24.2 Major mechanisms of DNA repair		
Mechanism	*Damage to DNA*	*DNA repair*
Base excision-repair	Damage to a single base due to spontaneous alteration or by chemical or radiation means.	Removal of the base by N–glycosylase; abasic sugar removal, replacement.
Nucleotide excision-repair	Damage to a segment of DNA by spontaneous, chemical or radiation means.	Removal of the DNA fragment (~ 30 nt length) and replacement.
Mismatch repair	Damage due to copying errors (1-5 base unpaired loops).	Removal of the strand (by exonuclease digestion) and replacement.
Double-strand break repair	Damage caused by ionizing radiations, free radicals, chemotherapy etc.	Unwinding, alignment and ligation.

1) BASE EXCISION-REPAIR:

The bases cytosine, adenine and guanine can undergo spontaneous depurination to respectively form uracil, hypoxanthine and xanthine. These altered bases do not exist in the normal DNA, and therefore need to be removed. This is carried out by base excision repair (Fig.24.16).

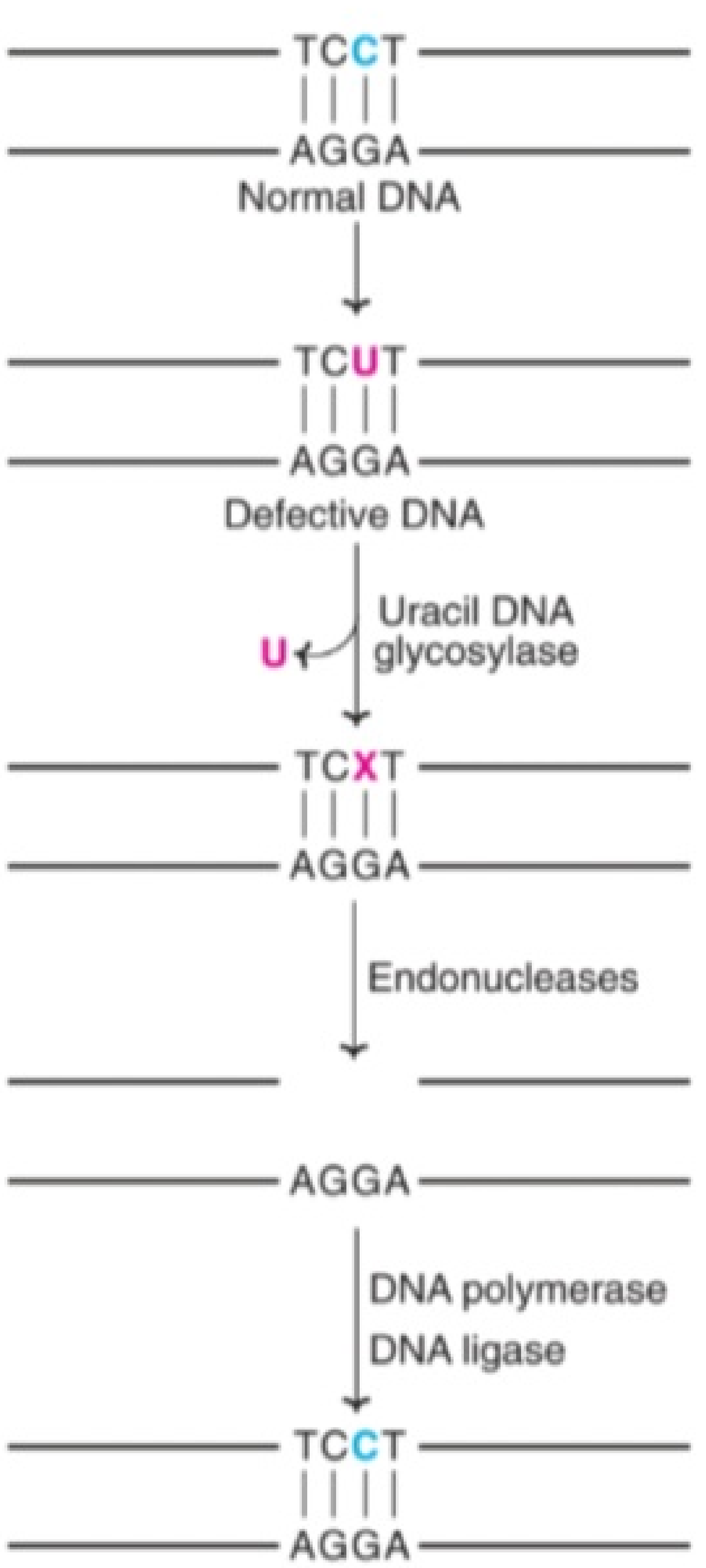

Fig. 24.16 : *A diagrammatic representation of base excision-repair of DNA.*

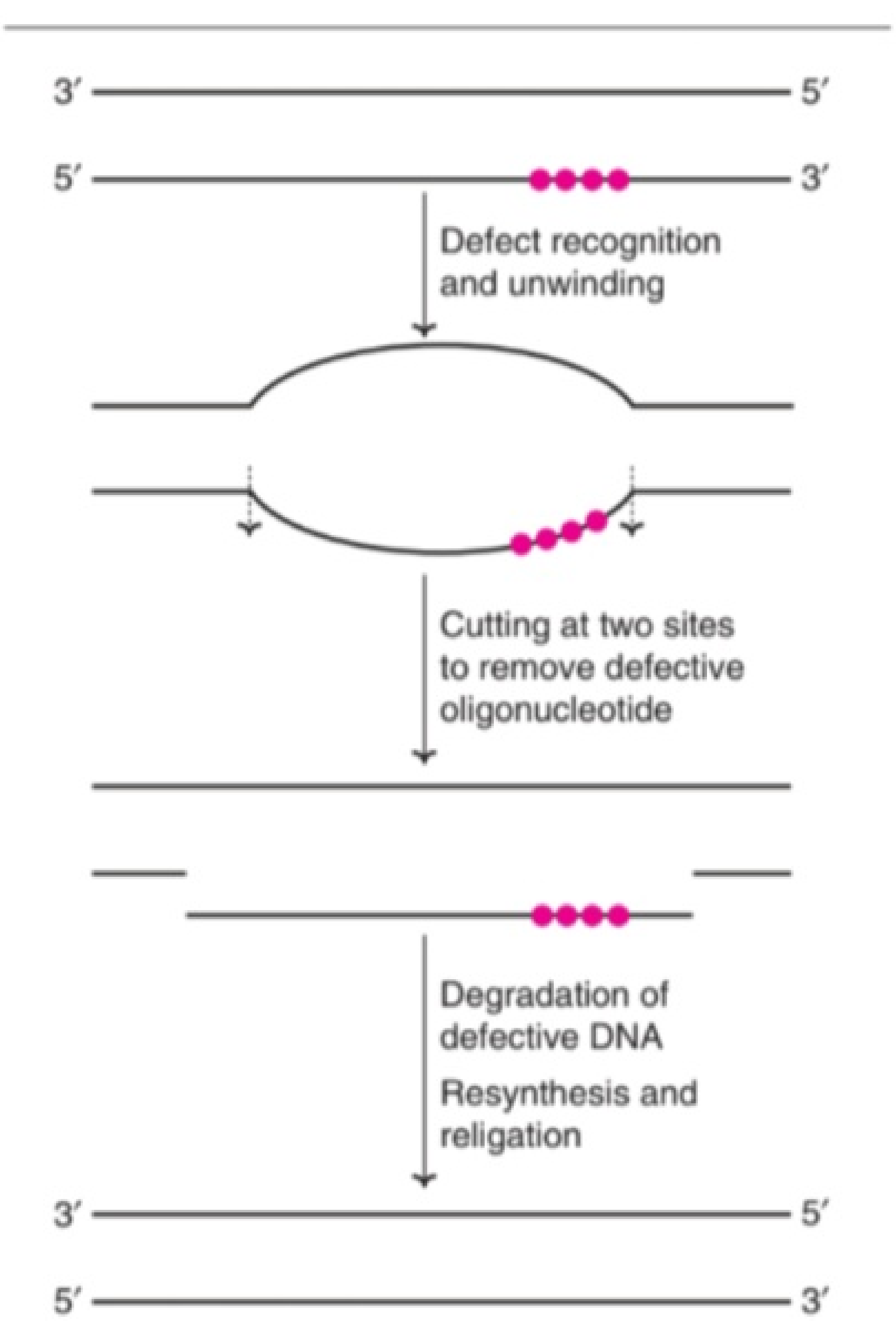

Fig. 24.17 : *A diagrammatic representation of nucleotide excision-repair of DNA.*

A defective DNA in which cytosine is deaminated to uracil is acted upon by the enzyme uracil DNA glycosylase. This results in the removal of the defective base uracil. An endonuclease cuts the backbone of DNA strand near the defect and removes a few bases. The gap so created is filled up by the action of repair DNA polymerase and DNA ligase.

2) NUCLEOTIDE EXCISION-REPAIR:

The DNA damage due to ultraviolet light, ionizing radiation and other environmental factors often results in the modification of certain bases, strand breaks, cross-linkages etc. Nucleotide excision-repair is ideally suited for such large-scale defects in DNA. After the identification of the defective piece of the DNA, the DNA double helix is unwound to expose the damaged part. An **excision nuclease** (exinuclease) cuts the DNA on either side (upstream and downstream) of the damaged DNA. This defective piece is degraded. The gap created by the nucleotide excision is filled up by DNA polymerase which gets ligated by DNA ligase (Fig.24.17).

Xeroderma pigmentosum (XP) is a rare autosomal recessive disease. The affected patients are photosensitive and susceptible to skin cancers. It is now recognized that XP is due to a defect in the nucleotide excision repair of the damaged DNA.

3) MISMATCH REPAIR:

Despite high accuracy in replication, defects do occur when the DNA is copied. For instance, cytosine (instead of thymine) could be incorporated opposite to adenine. Mismatch repair corrects a single mismatch base pair e.g. C to A, instead of T to A.

The template strand of the DNA exists in a methylated form, while the newly synthesized strand is not methylated. This difference allows the recognition of the new strands. The enzyme GATC endonuclease cuts the strand at an adjacent methylated GATC sequence (Fig.24.18). This is followed by an exonuclease digestion of the defective strand, and thus its removal. A new DNA strand is now synthesized to replace the damaged one.

Hereditary nonpolyposis colon cancer (HNPCC) is one of the most common inherited cancers. This cancer is now linked with **faulty mismatch repair** of defective DNA.

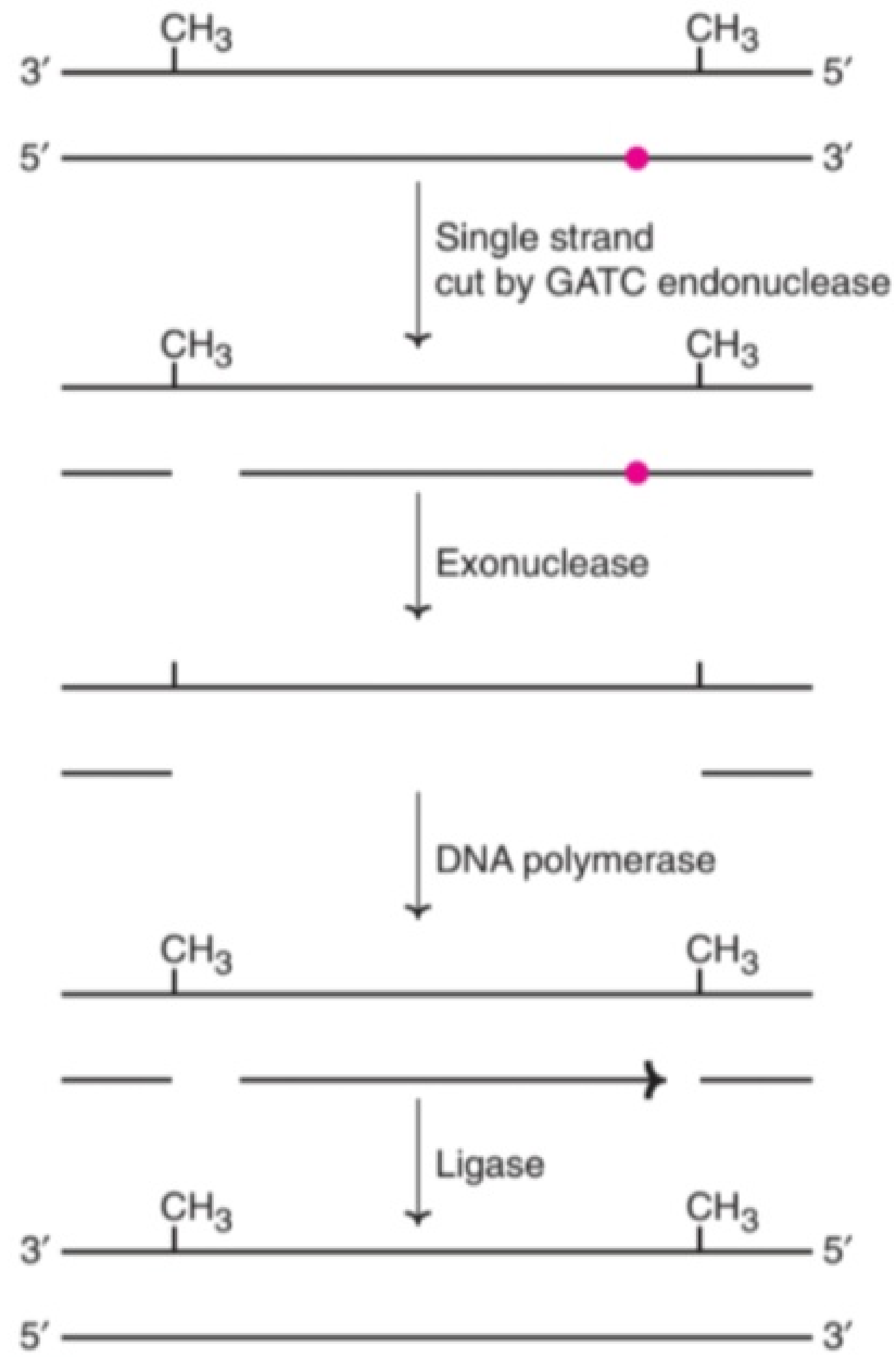

Fig. 24.18 : A diagrammatic representation of mismatch repair of DNA.

4) DOUBLE-STRAND BREAK REPAIR:

Double-strand breaks (DSBs) in DNA are dangerous. They result in genetic recombination which may lead to chromosomal translocation, broken chromosomes, and finally cell death. DSBs can be repaired by homologous recombination or non-homologous end joining. Homologous recombination occurs in yeasts while in mammals, non-homologous and joining dominates.

ADDITIONAL INFORMATION

Regulation of gene expression in presence of both glucose and lactose (Figure 21.2B).

When *E. coli* is exposed to both lactose and glucose, they first metabolize the glucose, although lactose is present. The cell does not induce those enzymes necessary for catabolism of lactose until the glucose has been exhausted.

- As glucose levels decrease, c-AMP levels increase. In the presence of c-AMP, a complex of c-AMP-**Catabolite activator Protein (CAP)** binds to the CAP binding site of promoter, stimulating the binding of the RNA polymerase to the promoter and transcription occurs.
- As glucose levels increase, c-AMP levels decrease and in the absence of c-AMP, CAP does not bind to CAP-binding site of promoter and transcription does not occur. Thus, the enzymes for metabolism of lactose are not produced if cells have an adequate supply of glucose even if the alternative energy source, lactose, is present at very high levels.

MUTATIONS

The term mutation refers to the permanent changes in the DNA sequence.

- Mutations in germ cells are transmitted to the next progeny and may give rise to inherited diseases.
- Mutations in somatic cells are not transmitted to the progeny but are important in the causation of cancers and some congenital malfunctions.

Causes of Mutations

Mutation arises by a number of different means:

- **Errors in replication**, If a mismatch base pair is not corrected during proofreading and post replication repair system.
- **Error due to recombination events.**
- **Spontaneous change in DNA**, e.g. deamination of cytosine to uracil or spontaneous depurination.
- **Environmental factors** like chemical mutagens and irradiations, e.g. UV light or ionizing radiation can alter the structure of DNA.

Types of Mutations (Figure 21.3)

Base Substitution or Point Mutation

- Point mutation occurs when only one base in DNA is altered, which may be transcribed into mRNA and therefore, may result in the translation of a protein with an abnormal amino acid sequence. Single base change can be of **transition type** or **transversion type**.
- **Transition,** in which one purine is replaced by another purine or one pyrimidine is replaced by another pyrimidine **(Figure 21.4)**.
- **Transversion,** in which a purine is replaced by a pyrimidine or a pyrimidine is replaced by a purine **(Figure 21.4)**.
- The point mutation can lead to:
  1. Silent mutation
  2. Missense mutation
  3. Nonsense mutation.

Silent mutation

Point mutations are said to be silent when there is no detectable effect. Silent mutation leads to the formation of a codon synonym and no change in the amino acid sequence of the protein occurs due to degeneracy of the codon, e.g. a codon change from CGA to CGG does not affect the proteins because both of these codons specify arginine.

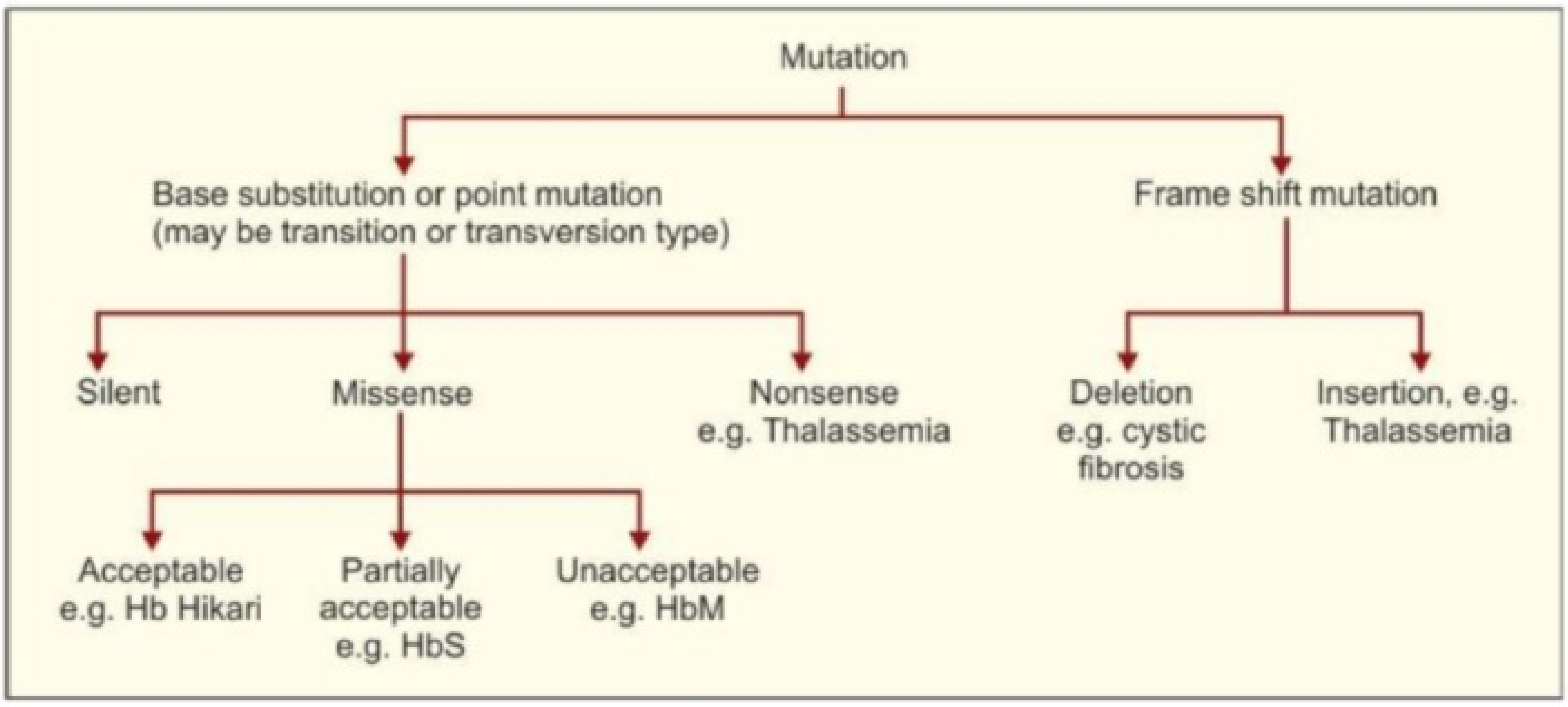

Figure 21.3: Types of mutation

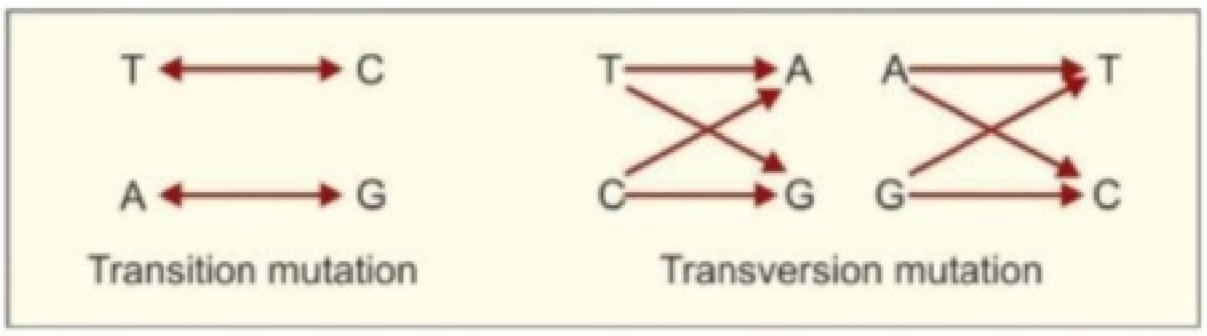

Figure 21.4: Diagrammatic representation of transition and transversion mutations

Missense mutation

Missense mutation will occur when a different amino acid is incorporated at the corresponding site in the protein molecule. Depending upon the location of the mistaken amino acid (missense) in the specific protein, missense mutation might be **acceptable, partially acceptable** or **unacceptable** with respect to the function of that protein.

Acceptable missense mutations

For example, ***Hb-Hikari***. This Hb has aspargine substituent for lysine at the 61 position in the β-globin chain. Hb-Hikari is a type of **transversion mutation,** in which either AAA or AAG changed to either AAU or AAC. The replacement of the specific ***lysine with aspargine*** does not alter the normal function of the β-chain in these individuals and is therefore called acceptable missense mutation.

Partially acceptable missense mutation

For example, HbS, ***sickle hemoglobin***, in which the normal amino acid in position 6 of the β-chain, glutamic acid has been replaced by valine. The corresponding transversion might be either GGA or GAG of ***glutamic acid*** *to* GUA or GUG of ***valine***.

Unacceptable missense mutation

For example, *Hb M (Methemoglobin)* in which normal amino acid in position 58 of α-chain, ***histidine*** has been replaced by ***tyrosine*** and nonfunctional Hb molecule is generated which cannot transport oxygen.

Nonsense mutations

Nonsense mutation leads to the conversion of an amino acid codon to a stop or nonsense codon. Nonsense mutation causes the premature termination of a polypeptide chain, which is usually nonfunctional, e.g. one type of ***thalassemia***, in which codon 17 of the β-chain is changed from UGG to UGA and results in the conversion of a codon tryptophan to a nonsense codon.

UGG ⟶ UGA
(Tryptophan) (Nonsense codon)

Frame Shift Mutations (Figure 21.5)

Frame shift mutation occurs when there is insertion or deletion of one or two nucleotides in DNA, that generates altered mRNAs.

Deletion frame shift mutation

- The deletion of a single nucleotide from the coding strand of a gene results in an altered reading frame in the mRNA.
- Since there is no punctuation in the reading of codons, the translating machinery does not recognize that a base was missing.
- Such frame shifts would result in the production of an entirely different protein after transcription and translation or indeed, no protein, at all, if a stop codon is encountered, as in cystic fibrosis of pancreas.

Insertion frame shift mutation

Insertion may be of one or two nucleotides. As with deletions, insertions of nucleotides into genes can lead to severe frame shift mutations, e.g. thalassemia.

If the number of nucleotides involved in deletion or insertion is three or multiples of three, frame shift does not occur. Instead, an abnormal protein, missing one or more amino acids, is synthesized. Such a mutation is likely to be less severe than a frame shift mutation.

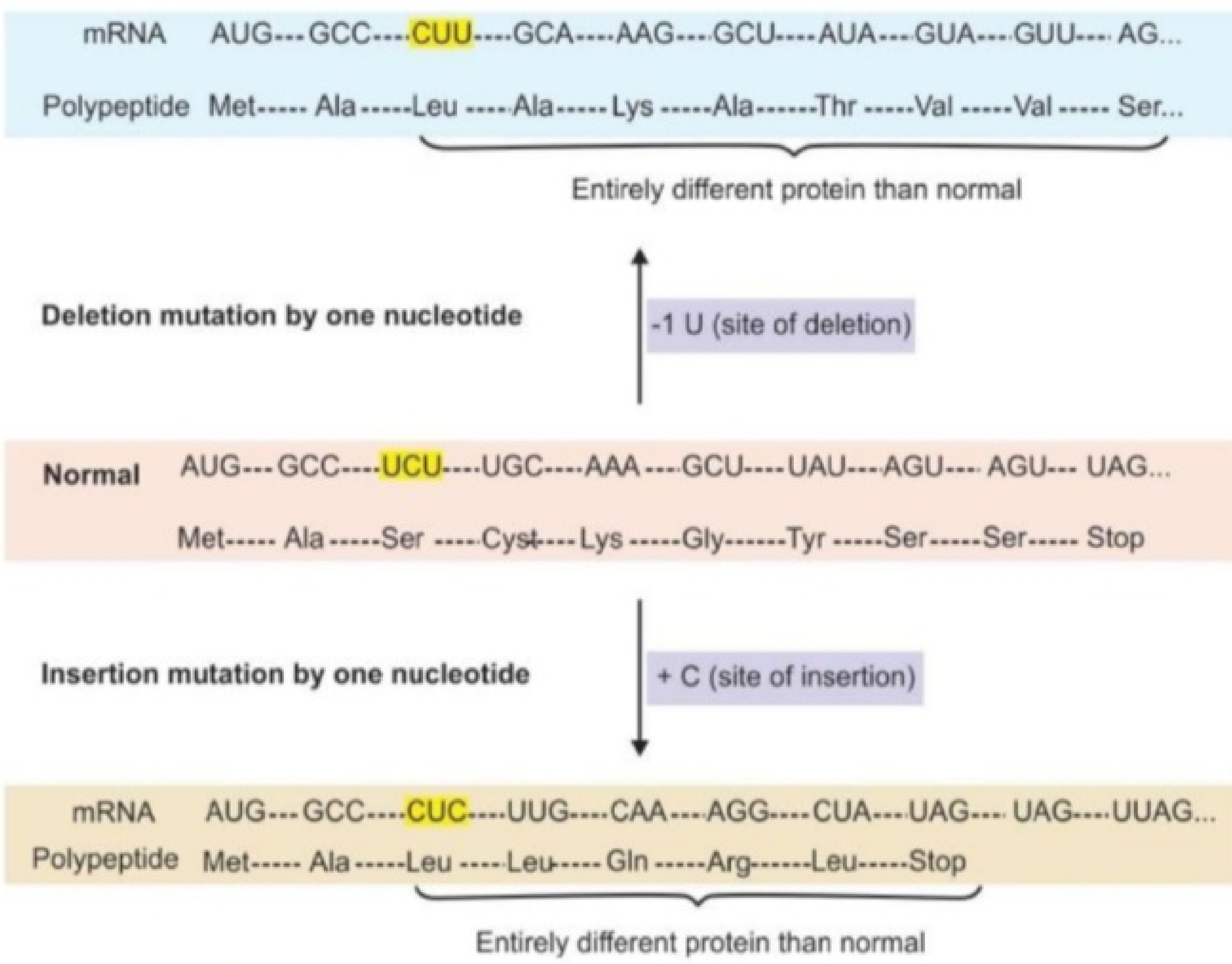

Figure 21.5: Diagrammatic representation of frame shift mutation

SUMMARY

- Mutations result when changes occur in the nucleotide sequence in DNA.
- Point mutations result from single base substitution. These may be transition or transversion type. Types of point mutations are silent mutation, missense mutation and nonsense mutation. Missense mutation might be acceptable or unacceptable to the function of that protein molecule.
- Frame shift mutations result from deletion or insertion of nucleotides in DNA that generates alterered mRNAs.
- An operon is a group of coordinately regulated genes. It consists of control sites (an operator and promoter) and a set of structural genes.
- The rate of expression of prokaryotic genes is controlled mainly at the level of transcription.